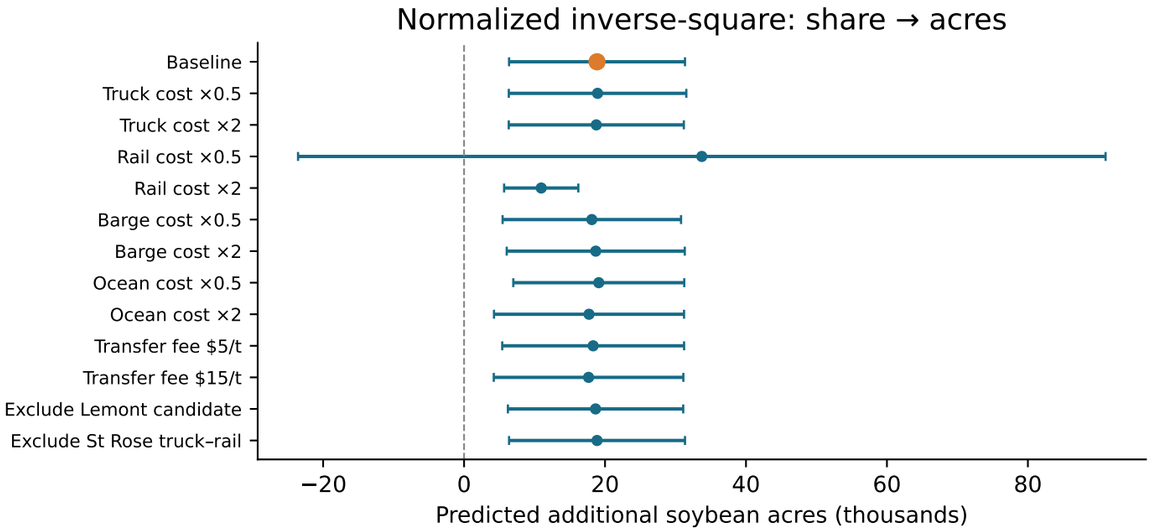
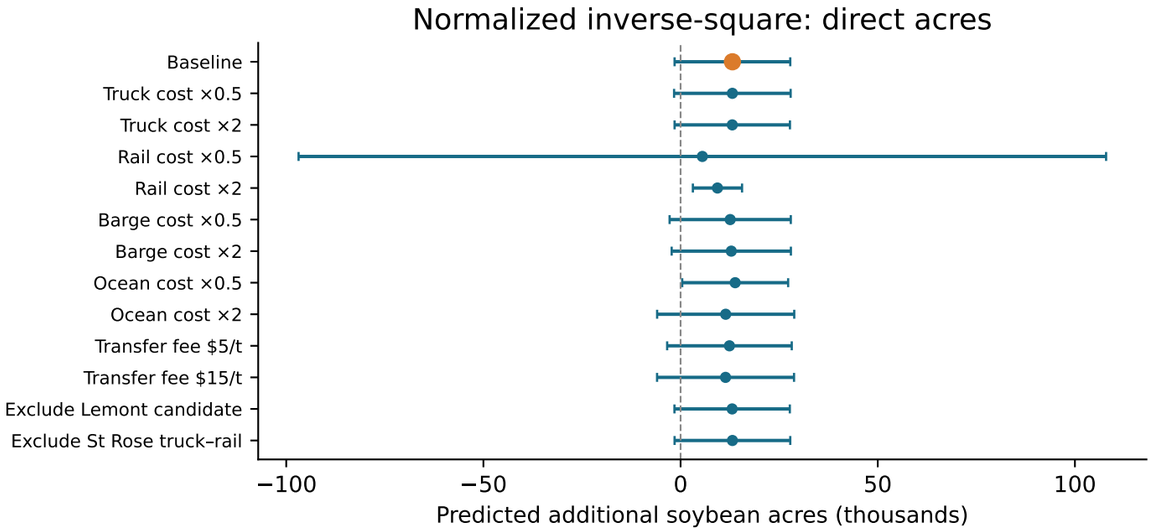
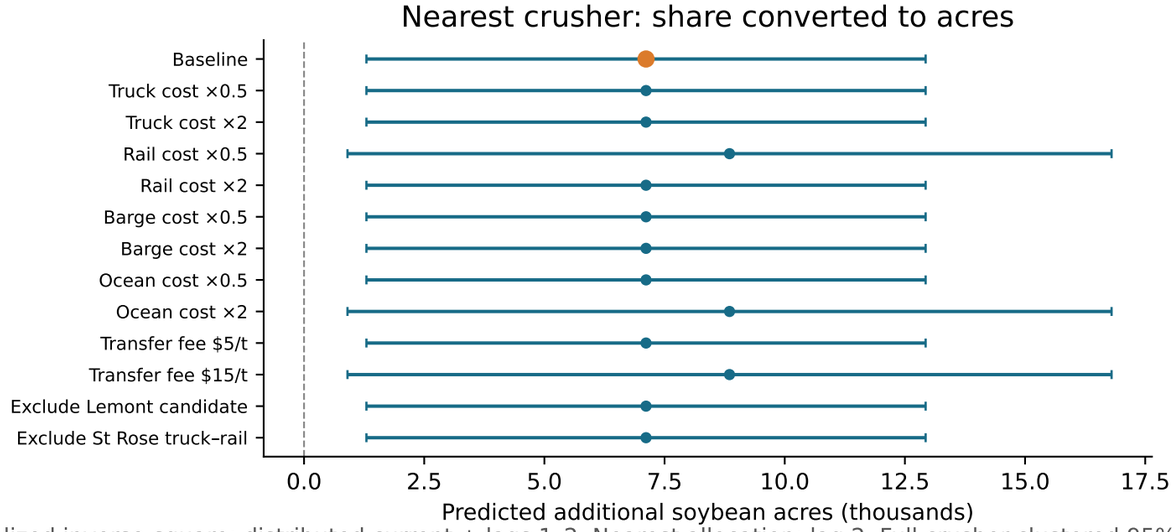
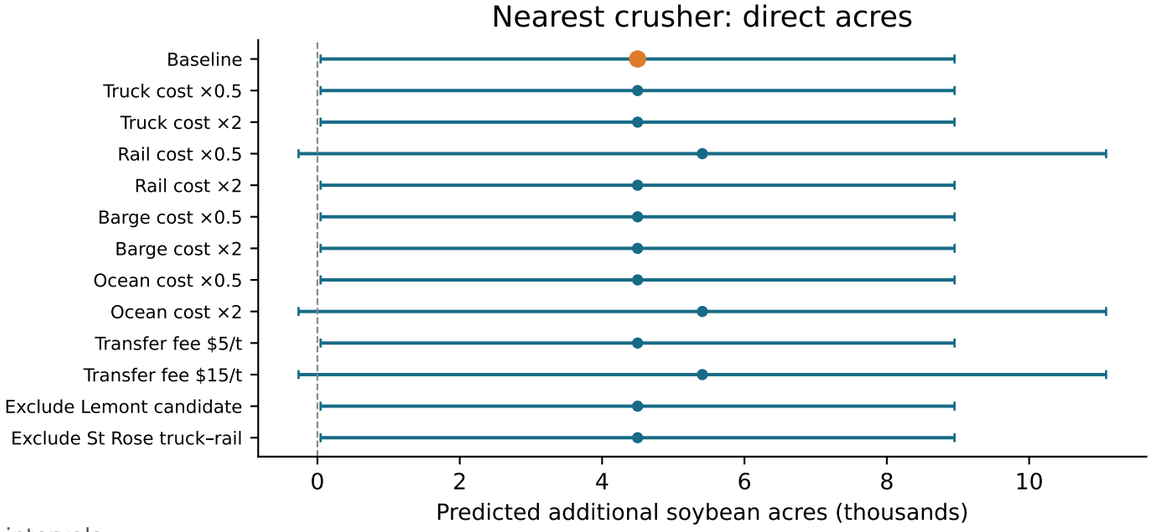
Transport-cost sensitivity: average +100 million gal/year, cumulative 0–75 miles









Normalized inverse-square: distributed current + lags 1–2. Nearest allocation: lag 2. Full crusher-clustered 95% intervals.
Modal multipliers and transfer fees are assumed stress tests. Routes reselected within documented-mode candidate sets.
Fixed baseline truck-cost unit and decay scales. Equal mean across 16 refinery sites. Multiple testing; observational estimates.